

Cluster & Regression + Tableau Introduction

MBA Study Notes

These notes focus on **business interpretation, managerial decision-making, use cases, assumptions, and Tableau-based business storytelling** - not heavy mathematical derivation or programming.

Each major topic includes a **case-based hands-on activity** using Excel and Tableau.

Learning Objectives

By the end of this module, you should be able to:

1. Explain how clustering supports business segmentation.
 2. Distinguish hierarchical and non-hierarchical clustering.
 3. Explain K-Means clustering using a business example.
 4. Interpret clusters from a managerial perspective.
 5. Understand linear and bivariate regression.
 6. Interpret regression coefficients and R^2 .
 7. Explain multiple regression and the role of multiple business drivers.
 8. Distinguish correlation from regression.
 9. Understand Tableau's basic components.
 10. Build a basic business dashboard and derive insights.
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Session Overview

Topic	Activity
Cluster Analysis	Customer Segmentation
Hierarchical Clustering	Store Segmentation + Dendrogram
K-Means	Customer Segmentation using K-Means
Linear/Bivariate Regression	Advertising vs Sales
Multiple Regression	Sales Prediction
Tableau Basics	Build Business Visualizations
Tableau Dashboard	Management Dashboard

PART 1 - Business Analytics & Clustering

Getting Started

Consider this business question:

"If you are the marketing manager of an e-commerce company with 100,000 customers, should you send the same marketing campaign to everyone?"

Think about why customers may need to be treated differently based on:

- Income levels
- Spending patterns
- Purchase frequencies
- Product preferences
- Customer lifetime values
- Price sensitivities

This need for differentiation leads to **customer segmentation** and **clustering**.

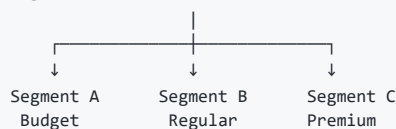
1. What is Clustering?

Clustering is an **unsupervised** analytical technique used to group similar observations together.

Unlike classification, we don't start with predefined labels.

Example:

CUSTOMERS



The algorithm identifies groups based on similarities in the available data.

Business Applications

Marketing

- Customer segmentation

Banking

Segment customers based on income, transactions, balance, and credit behaviour.

Retail

Group stores based on revenue, footfall, and product mix.

HR

Identify employee groups based on experience, performance, compensation, and training participation.

Supply Chain

Group suppliers based on cost, delivery time, quality, and reliability.

2. Clustering vs Classification

Clustering	Classification
Unsupervised	Supervised
No predefined target	Has predefined target
Discovers groups	Predicts known categories
Customer segmentation	Churn prediction
Market segmentation	Loan approval prediction

Business example

- If customer categories (Premium / Standard / Basic) are known and you want to **predict** the category → **Classification**
 - If categories are unknown and you want the algorithm to **discover** natural groups → **Clustering**
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3. Clustering Example - Retail

Suppose a retailer has this information:

Customer	Income	Annual Spend	Visits
A	30K	8K	5
B	32K	10K	6
C	90K	70K	25
D	85K	65K	23
E	55K	35K	15

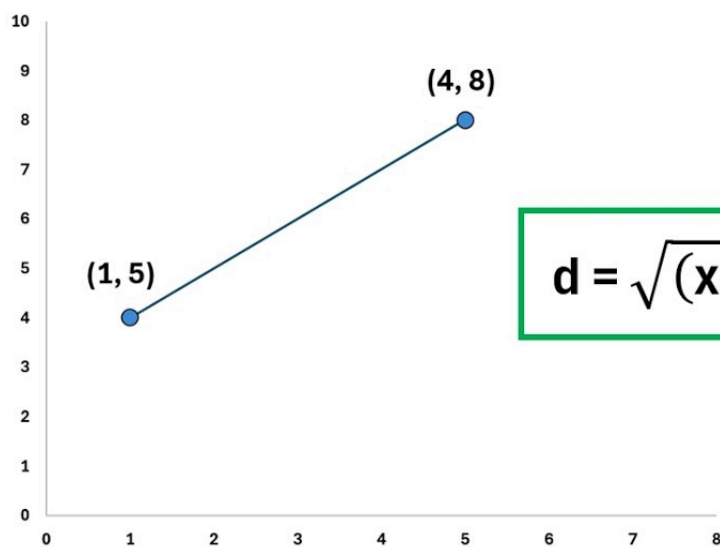
Reflection: Based on the data above, what customer segments can you identify? What business actions might follow from each segment?

4. Similarity and Distance

Clustering requires a measure of similarity/dissimilarity.

One common measure is **Euclidean distance**:

Euclidean Distance



Key idea:

- Smaller distance → more similar observations
 - Larger distance → less similar observations
-

5. Why Data Scaling Matters

Consider:

- **Age** → 20–60
- **Income** → ₹20,000–₹2,00,000

Income has a much larger numerical scale and may dominate distance-based algorithms.

Analysts often **standardize** variables before clustering:

$$Z = \frac{X - \mu}{\sigma}$$

Standardization puts variables on comparable scales.

Hands-on Activity - Customer Segmentation

Business Scenario

You are a Marketing Manager for an online retail company. The company wants to identify different customer segments so that marketing campaigns can be customized.

Dataset

Customer	Age	Income ₹000	Annual Spend ₹000	Visits/Year
C01	23	25	8	5
C02	25	28	10	6
C03	27	30	12	7
C04	45	55	35	15
C05	48	60	40	17
C06	42	52	32	14
C07	35	80	65	22
C08	38	85	70	25
C09	40	90	75	27
C10	29	40	20	10

Your Tasks

Step 1 - Identify segmentation variables

Which variables would you use for segmentation? Should Customer ID be used? Why or why not?

Step 2 - Initial hypothesis

Which customers appear similar even before running a clustering algorithm?

Step 3 - Scatter plot

Create a scatter plot of **Income vs Annual Spend**.

- Do you see natural groups?
- Which customers appear similar?
- Are there any unusual observations?

Step 4 - Business interpretation

Suppose three clusters are identified. Complete the table below:

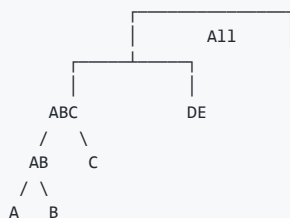
Cluster	Income	Spend	Visits	Business Profile	Marketing Strategy
Cluster 1					
Cluster 2					
Cluster 3					

Reflection: What marketing strategy would you recommend for each segment?

PART 2 - Hierarchical Clustering

6. What is Hierarchical Clustering?

Hierarchical clustering creates a hierarchy of groups, often shown as a **dendrogram**:



This shows how observations/clusters are progressively merged or divided.

7. Agglomerative Clustering

A **bottom-up** approach:

A B C D E → AB C DE → ABC DE → ABCDE

Each observation starts as its own cluster; similar observations are then merged.

8. Divisive Clustering

A **top-down** approach:

ABCDE → ABC DE → (continue splitting)

9. Dendrogram Interpretation

If you cut a dendrogram at a particular level and get four branches, what does that mean?

The dendrogram is useful for **exploratory analysis** and understanding hierarchical relationships between groups.

10. Business Example - Store Segmentation

A retail chain with 20 stores might use variables such as:

- Revenue
- Footfall
- Average transaction value
- Product diversity

Hierarchical clustering can help identify similar stores. Management can then investigate why stores in different groups behave differently.

11. Advantages & Limitations

Advantages

- Provides a visual hierarchy
- Useful for exploratory analysis
- Does not require fixing the number of clusters at the start
- Helps understand relationships between groups

Limitations

- Can be computationally expensive for large datasets
 - Sensitive to distance and linkage choices
 - Early merging decisions can affect the final hierarchy
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Hands-on Activity - Store Segmentation

Business Scenario

A retail company operates 8 stores. Management wants to understand which stores behave similarly.

Dataset

Store	Revenue ₹L	Footfall 000	Avg Transaction ₹
S1	50	20	2500
S2	55	22	2500
S3	30	25	1200
S4	32	27	1185
S5	80	25	3200
S6	85	27	3150
S7	20	15	1333
S8	22	16	1375

Your Tasks

1. Which stores appear similar?
 2. Which stores appear very different?
 3. If hierarchical clustering is applied, which stores might merge first?
 4. Generate a dendrogram (if using supporting software). If management wants 3 store groups, where would you cut the dendrogram?
 5. Should management use the same strategy for all store groups? What strategy would you recommend for each group?
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PART 3 - Non-Hierarchical Clustering (K-Means)

12. What is K-Means?

K-Means divides observations into **K** groups based on distance to cluster centers.

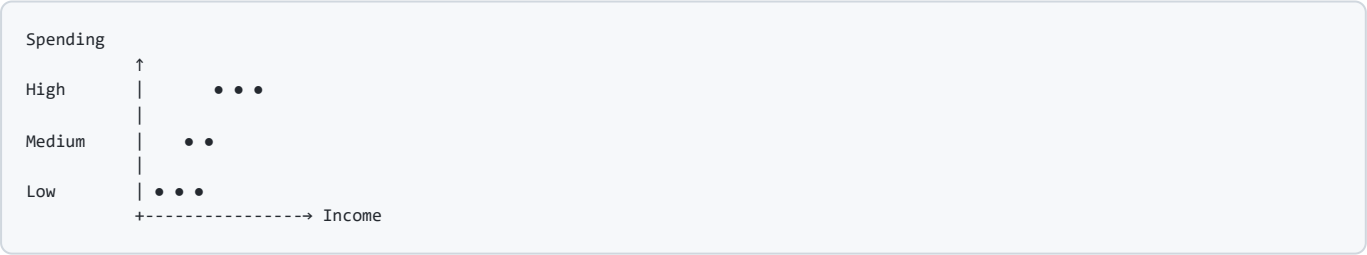
If $K = 3$, the algorithm produces three clusters.

13. K-Means Process

Step	Action
1	Choose K
2	Initialize centroids
3	Assign each observation to the closest centroid
4	Recalculate cluster centers
5	Reassign observations
6	Repeat until convergence

14. Simple Business Example

For customers with annual income and spending score, K-Means might produce:



The resulting clusters can then be profiled for business action.

15. Choosing K - Elbow Method

K	Within-cluster variation
1	900
2	500
3	250
4	220
5	205

The point where improvement becomes relatively small is called the **elbow**.

When choosing K, use both quantitative evidence and business interpretability.

16. Hierarchical vs Non-Hierarchical

	Hierarchical	K-Means
Approach	Builds hierarchy	Creates partitions
Common output	Dendrogram	Cluster assignments
K required	Not necessarily initially	Yes
Visualization	Strong	Strong with plots
Large datasets	Can be expensive	Generally more scalable
Centroids	Not central to method	Central to method
Use	Exploration	Segmentation/partitioning

Hands-on Activity - Mall Customer Segmentation

Dataset

Customer	Income ₹000	Spending Score
C1	20	15
C2	25	20
C3	30	25
C4	35	30
C5	45	45
C6	50	50
C7	55	55
C8	60	60
C9	75	80
C10	80	85
C11	85	90
C12	90	95

Your Tasks

1. Create a scatter plot with **Income** on the X-axis and **Spending Score** on the Y-axis.
2. How many customer segments would you create as a marketing manager?
3. Apply K-Means with **K = 2**, **K = 3**, and **K = 4**. What changes when K changes?
4. After choosing an appropriate K, complete the table:

Segment	Characteristics	Recommended Strategy

PART 4 - Regression

Getting Started

“If advertising expenditure increases, can we estimate how sales might change?”

This question introduces **regression**.

17. What is Regression?

Regression models the relationship between a **dependent variable** and one or more **predictor variables**.

Predictor	Outcome
Advertising	Sales
Price	Demand
Experience	Salary
Discount	Revenue
Area	House Price

18. Simple Linear Regression

Simple Linear Regression



$$\hat{y} = b_0 + b_1 X_1$$

Diagram illustrating the components of the Simple Linear Regression equation:

- $\hat{y}$: Dependent variable
- b_0 : y-intercept (constant)
- b_1 : Slope coefficient
- X_1 : Independent variable

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- Y = dependent variable
- X = predictor
- b_0 = intercept
- b_1 = slope/coefficient

19. Illustrative Example

Sales = 100 + 5(Advertising)

If advertising = 10: Sales = 100 + (5 * 10) = 150

20. Interpreting the Slope

In the equation $\text{Sales} = 100 + 5(\text{Advertising})$, the coefficient 5 indicates that a one-unit increase in advertising is associated with an estimated 5-unit increase in predicted sales (depending on the units used).

21. Intercept

The intercept ($b_0 = 100$) is the predicted value of Y when $X = 0$.

Note: An intercept may not be meaningful if $X = 0$ is unrealistic or outside the observed data range.

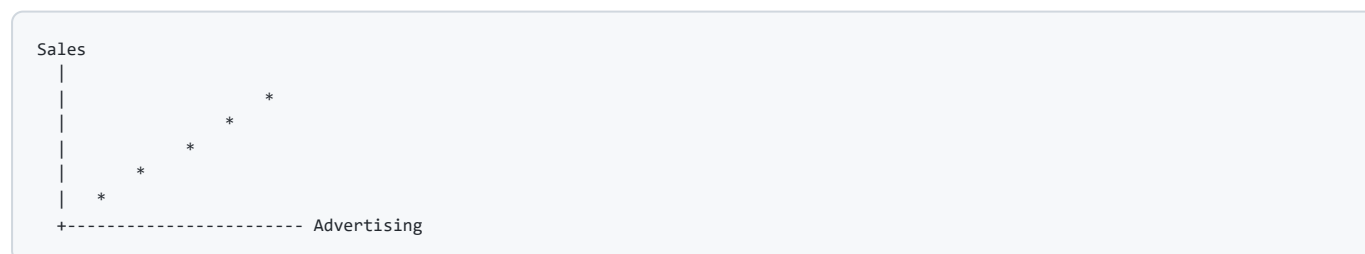
22. Bivariate Regression

Bivariate regression examines the relationship between two variables - one predictor and one outcome.

Example: Advertising Spend → Sales

Advertising	Sales
10	100
20	120
30	145
40	160
50	185

23. Scatter Plot



- Upward movement → positive association
 - Downward movement → negative association
-

24. Correlation

$-1 \leq r \leq 1$

r value	Meaning
$r \approx +1$	Strong positive linear association
$r \approx 0$	Little/no linear association
$r \approx -1$	Strong negative linear association

25. Correlation Does Not Mean Causation

Two variables may move together without one causing the other. A third factor can influence both.

Correlation indicates association, not automatically causation.

26. Residual

Residual = Actual - Predicted

Example: Actual = ₹150 lakh, Predicted = ₹145 lakh → Residual = 5

27. R^2

$R^2 = 0.75$ means the model accounts for approximately 75% of variation in the dependent variable.

R^2 is not proof of causality and does not guarantee good predictions on new data.

Hands-on Activity - Advertising and Sales

Business Scenario

You are a Business Analyst for a consumer goods company. The marketing manager wants to know:

"Does advertising expenditure have a relationship with sales?"

Dataset

Month	Advertising ₹L	Sales ₹L
Jan	5	52
Feb	7	58
Mar	8	61
Apr	10	68
May	12	72
Jun	14	78
Jul	15	81
Aug	18	89
Sep	20	94
Oct	22	101

Your Tasks

Task 1 - Scatter Plot

Plot Advertising (X-axis) vs Sales (Y-axis). What pattern do you observe?

Task 2 - Correlation

Calculate the correlation coefficient.

- Is the relationship positive or negative?
- Is it weak or strong?

Task 3 - Regression Line

Fit: Sales = b_0 + b_1 Advertising

Identify the intercept, slope, and R^2 .

Task 4 - Prediction

If advertising expenditure is ₹25 lakh, what sales does your regression model predict?

Task 5 - Reflection

Can the marketing manager conclude that increasing advertising will definitely increase sales? What other factors might influence sales?

PART 5 - Multiple Regression

“Can sales depend only on advertising?”

Sales may also depend on price, discount, season, competitor activity, distribution, and brand awareness. This leads to **multiple regression**.

28. Multiple Regression

$$Y_i = b_0 + b_1X_{1i} + b_2X_{2i} + b_nX_{ni} + u_i$$

Y_i = dependent variable

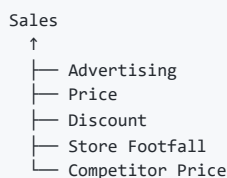
b_0 = Intercept

$b_1 \dots b_n$ = Coefficient of Regression

$X_{1i} \dots X_{ni}$ = independent variable

u_i = disturbance error

29. Business Example - Sales



Sales = b_0 + b_1 Advertising + b_2 Price + b_3 Discount + b_4 Footfall

30. Interpreting Multiple Regression

In a model such as:

Sales = 20 + 4(Advertising) + 2(Discount) + 0.5(Footfall)

The advertising coefficient is interpreted **holding other predictors constant**.

31. Why Multiple Regression Matters

Multiple regression helps answer questions such as:

- What factors are associated with sales?
 - Which variables have stronger relationships with the outcome?
 - How does price relate to demand after accounting for other variables?
 - How accurately can sales be predicted?
-

32. Multicollinearity

When predictors are strongly related to each other (e.g., house area, bedrooms, and rooms), coefficients can become unstable and individual interpretation becomes difficult. This is called **multicollinearity**.

33. Model Evaluation

Metric	Formula	Notes
MAE	$MAE = \frac{1}{n} \sum_{i=1}^n y_i - \hat{y}_i $ Σ sum of n is the number of data points y_i actual value at the i-th data point $\hat{y}$ predicted value at the i-th data point	Easy to interpret in target units
MSE	$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$ $\frac{1}{N}$ average over all results y_i true y $\hat{y}_i$ estimate of y 2 makes result quadratic	Large errors weighted more
RMSE	$RMSE = \sqrt{\sum_{i=1}^n \frac{(\hat{y}_i - y_i)^2}{n}}$	Same units as dependent variable

Hands-on Activity - What Drives Sales?

Business Scenario

"Advertising is not the only factor affecting sales. I want to understand several factors simultaneously."

Dataset

Store	Advertising	Price	Discount %	Footfall	Sales
A	5	100	5	500	60
B	8	95	10	600	72
C	10	90	15	700	82
D	12	105	10	750	88
E	15	95	20	850	105
F	18	90	15	900	115
G	20	85	25	1000	130
H	22	80	20	1100	140

Your Tasks

Build a multiple regression model:

$$\text{Sales} = b_0 + b_1 \text{Advertising} + b_2 \text{Price} + b_3 \text{Discount} + b_4 \text{Footfall}$$

Then interpret your results:

1. Which variables have positive coefficients?
2. Which variables have negative coefficients?
3. Which variables appear to have stronger relationships with sales?
4. What is the R^2 ?
5. How well does the model explain variation in sales?
6. Interpret the statement: "Advertising has a coefficient of 2.5."

Managerial Decision Case

A company's regression model shows that discount percentage is positively associated with sales, but management is concerned that excessive discounting could reduce profitability.

Question 1: Should management simply increase discounts?

Question 2: What additional information would you need to make a recommendation?

PART 6 - Tableau Basics

34. What is Tableau?

Tableau is a business intelligence and visualization platform used to:

- Connect to datasets
- Explore data
- Create visualizations
- Build dashboards
- Apply filters
- Communicate business insights

35. Tableau Workflow

Connect Data → Understand Data → Create Worksheet → Build Visualization
→ Combine Worksheets → Create Dashboard → Add Filters → Generate Business Insights

36. Important Tableau Terminology

Term	Description
Workbook	Container for Tableau analysis
Worksheet	Individual visualization
Dashboard	Combination of multiple worksheets
Story	Sequence of visualizations for narrative

37. Dimensions vs Measures

Type	Description	Examples
Dimensions	Categorical fields	Region, Category, Product, State
Measures	Numerical, aggregatable	Sales, Profit, Quantity, Revenue

38. Example Sales Dataset

Order Date	Region	Category	Sales	Profit	Quantity
Jan	North	Furniture	50000	6000	120
Jan	South	Technology	70000	14000	90
Feb	East	Furniture	45000	4000	100
Feb	West	Technology	80000	18000	110
Mar	North	Office	35000	7000	150
Mar	South	Furniture	55000	5000	130
Apr	East	Technology	90000	20000	125
Apr	West	Office	40000	8000	160

Dimensions: Region, Category, Order Date

Measures: Sales, Profit, Quantity

39. Basic Tableau Charts

Chart Type	Best For	Example
Bar Chart	Comparing categories	Sales by Region
Line Chart	Trends over time	Monthly Sales
Scatter Plot	Relationships between variables	Sales vs Profit
Map	Geographic analysis	Sales by State

40. Filters

Filters allow interactive focus on subsets of data (e.g., Region = South, Category = Technology).

One interactive dashboard can replace separate reports for every region.

PART 7 - Tableau Hands-On Activity

Case: Retail Sales Analysis

“You are a business analyst supporting the regional sales manager. Build a dashboard that helps the manager understand sales and profitability.”

Use the dataset from **Section 38**.

Activity 1 - Sales by Region

Field	Placement
Region	Rows
Sales	Columns

Questions:

1. Which region has the highest sales?
 2. Which has the lowest?
 3. Is sales distribution balanced?
-

Activity 2 - Sales Trend

Field	Placement
Order Date	Columns
Sales	Rows

Create a **line chart** and answer:

- What is the trend?
 - Which month has the highest sales?
 - Is there evidence of seasonality?
-

Activity 3 - Profit by Category

Field	Placement
Category	Rows
Profit	Columns

Questions:

- Which category is most profitable?
 - Is the category with the highest sales also the most profitable?
-

Activity 4 - Sales vs Profit

Field	Placement
Sales	Columns
Profit	Rows

Question: Do higher sales generally correspond to higher profit? Identify any outliers.

Activity 5 - Mini Dashboard

Build a dashboard with this layout:

SALES PERFORMANCE		
Total Sales	Total Profit	Total Quantity
SALES TREND		
Sales by Region	Profit by Category	
Sales vs Profit		

Filters: Region | Category | Date

Business Discussion Questions

After building your dashboard, reflect on:

- 1. If Region A has the highest sales but relatively low profit, what would you investigate?
- 2. If sales are increasing but profit is decreasing, what could be happening?
- 3. Should management automatically invest more in the region with the highest sales?

Group Activity - E-Commerce Customer Analytics

“You are a consulting team hired by an e-commerce company. Management wants to understand customer segments, identify factors associated with spending, and create a dashboard for senior management.”

Dataset Fields

- Customer ID
- Age
- Income
- Annual Spend
- Purchase Frequency
- Discount Usage
- Website Visits
- Region

Team Tasks

Task	Activity
Task 1 - Clustering	Identify customer segments
Task 2 - Regression	Model Annual Spend using Income, Purchase Frequency, Website Visits, Discount Usage
Task 3 - Tableau	Create customer count, average spend, customers by segment, income vs spend, and regional distribution
Task 4 - Presentation	Present 3 findings and 2 business recommendations or questions for further investigation

Worksheet Template

Cluster Analysis

- Variables selected:
- Number of clusters:
- How was the number selected?
- Cluster 1 characteristics:
- Cluster 2 characteristics:
- Cluster 3 characteristics:
- Business interpretation:

Regression

- Dependent variable:
- Predictors:
- Regression equation/model:
- R^2 :
- Important coefficient:
- Interpretation:

Tableau

- Visualization 1:
- Visualization 2:
- Visualization 3:
- Key insight:

Management Recommendations

- Finding 1:
- Finding 2:
- Finding 3:
- Business action / question for further investigation:

Summary

Cluster Analysis

Discover natural groups in data.

Type	Output / Method
Hierarchical	Dendrogram
Non-Hierarchical	K-Means partitions

Regression

Model relationships and estimate/predict a numerical outcome.

Type	Description
Simple/Bivariate	One predictor → One outcome
Multiple Regression	Multiple predictors → One outcome

Tableau

Convert data into interactive visual insights.

Data → Worksheet → Visualization → Dashboard → Filters → Business Insight

Key Concepts

Concept	Business Question
Clustering	Who/what belongs together?
Hierarchical clustering	How are groups related hierarchically?
K-Means	How can we segment observations into K groups?
Regression	How is an outcome related to one or more predictors?
Multiple regression	How do several factors relate to an outcome simultaneously?
R ²	How much variation does the model account for?
Tableau	How can we communicate the analysis visually?
Dashboard	What should a manager see and act upon?

The Business Analytics Flow

Step	Role
Clustering	Segment the business
Regression	Understand relationships and make estimates
Tableau	Communicate the findings

data → analysis → interpretation → business decision